

Candidate Name

Centre Number

Candidate Number

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL****General Certificate of Education Advanced Level****CHEMISTRY****9189/5**

PAPER 5 Practical Test

NOVEMBER 2013 SESSION

1 hour 20 minutes

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Insert

TIME 1 hour 20 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 5 printed pages and 3 blank pages.

Copyright: Zimbabwe School Examinations Council, N2013.

- 1 **FA1** is a sample of impure water, suspected to be containing halide ions.

FA2 is $0.005 \text{ mol dm}^{-3} \text{ AgNO}_{3(\text{aq})}$.

You are required to determine the concentration of the halide ions present in **FA1**.

Using a burette place between 28.20 cm^3 and 28.80 cm^3 of **FA1** into a 250 cm^3 volumetric flask.

Make it up to the mark using distilled water and label it **FA3**. Record your results in the **Dilution Table**.

Dilution table

final burette reading / cm^3	
initial burette reading / cm^3	
volume of FA1 used / cm^3	

[3]

Pipette 25 cm^3 of **FA3** into a conical flask and add four drops of $\text{K}_2\text{CrO}_{4(\text{aq})}$ indicator.

Titrate against **FA2** until the first permanent colour change.

Record your results in the **Titration Table**.

Repeat the titration as many times as you think necessary to obtain accurate results.

Titration Table

titration number					
final burette reading / cm^3					
initial burette reading / cm^3					
volume of FA2 used					

[15]

Summary

25 cm^3 of **FA3** required _____ cm^3 **FA2** for complete precipitation.

- (a) Calculate the number of moles of $\text{Ag}^+_{(\text{aq})}$ in the titre given that the concentration of $\text{AgNO}_{3(\text{aq})}$ is $0.005 \text{ mol dm}^{-3}$.

[1]

- (b) (i) Assuming the halide ions present to be Cl^- and I^- , calculate the total number of moles of anions in the volume of impure water used.

[4]

- (ii) Calculate the concentration of the halide ions in the impure water.

[3]

- (c) Suggest a possible source of error in this experiment.

[1]

ASSESSMENT OF PLANNING SKILLS

For
Examiner's
Use

DO NOT CARRY OUT YOUR PLAN

- 2 You are required to design an experiment to show how you can use the results in **question 1** to estimate the proportions of Cl^- and I^- in **FA1**.

Assume you are provided with the following:

filter paper and funnel

balance

burette

beaker

conical flask

dilute $\text{NH}_3(\text{aq})$

State in a sequence of numbered steps how you would

1. perform the experiment,
2. determine the proportions of Cl^- and I^- in **FA1**.

For
Examiner's
Use

[13]